

UNIT IV :--HTML 5

4.1 Introduction to HTML 5

4.2 Advantage and Disadvantages

4.3 Elements in HTML 5

4.1 Introduction to HTML 5

HTML5 tutorial provides details of all 40+ HTML tags including audio, video, header, footer, data, datalist, article etc. This HTML tutorial is designed for beginners and professionals.

HTML5 is a next version of HTML. Here, you will get some brand new features which will make HTML much easier. These new introducing features make your website layout clearer to both website designers and users. There are some elements like <header>, <footer>, <nav> and <article> that define the layout of a website.

Why use HTML5

It is enriched with advance features which makes it easy and interactive for designer/developer and users.

It allows you to play a video and audio file.

It allows you to draw on a canvas.

It facilitate you to design better forms and build web applications that work offline.

It provides you advance features for that you would normally have to write JavaScript to do.

Since HTML5 is relatively new hence it can run only on modern browsers like Chrome (version 61 and above) and older browsers like IE will face difficulties while loading HTML5 content. Its ability to run on different sets of devices makes handling the responsiveness of the websites on different devices a headache.

In HTML5 most of the code is rendered/cached on the client-side so it becomes easy for the user to access the entire code and use of local storage to store any crucial information about the site or use is not a good idea because it is also easily accessible and can be vulnerable.

HTML5 made it possible to include different formats of media, and according to the devices, you have to include multiple formats of the same media and that will cause an increase in the media licensing cost

HTML 5 Example

Let's see a simple example of HTML5.

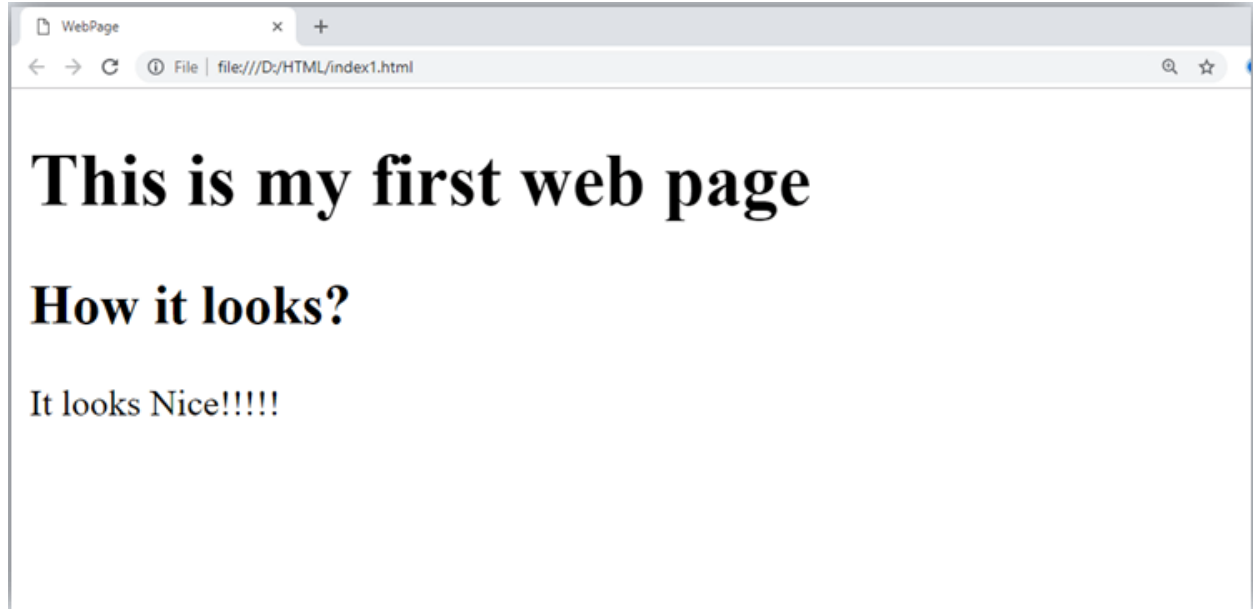
```
<!DOCTYPE>
<html>
<body>
<h1>Write Your First Heading</h1>
<p>Write Your First Paragraph.</p>
</body>
</html>
```

HTML Elements

An HTML file is made of elements. These elements are responsible for creating web pages and define content in that webpage. An element in HTML usually consist of a start tag <tag name>, close tag </tag name> and content inserted between them. **Technically, an element is a collection of start tag, attributes, end tag, content between them.**

```
<html>
<head>
  <title>WebPage</title>
</head>
<body>
  <h1>This is my first web page</h1>
  <h2> How it looks?</h2>
  <p>It looks Nice!!!!!!</p>
```

```
</body>
</html>
```



Block-level and Inline HTML elements

For the default display and styling purpose in HTML, all the elements are divided into two categories:

Block-level element:

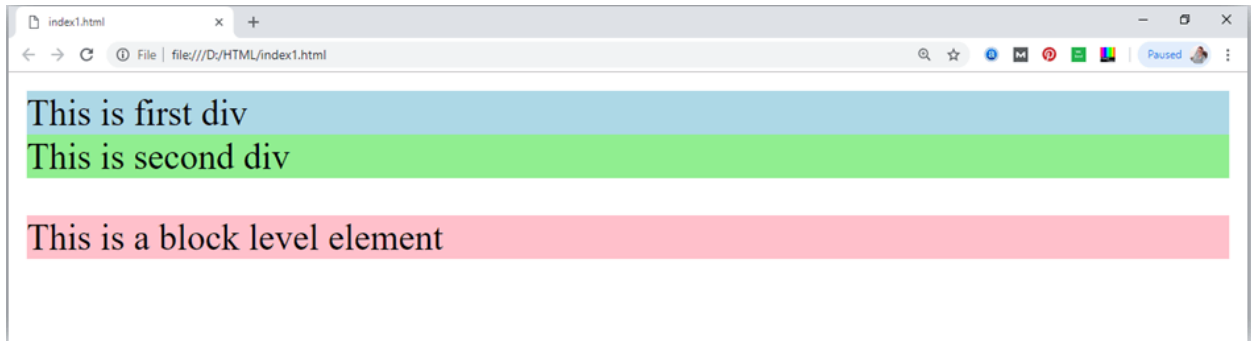
- These are the elements, which structure main part of web page, by dividing a page into coherent blocks.
- A block-level element always start with new line and takes the full width of web page, from left to right.
- These elements can contain block-level as well as inline elements.

Following are the block-level elements in HTML.

<address>, <article>, <aside>, <blockquote>, <canvas>, <dd>, <div>, <dl>, <dt>, <fieldset>, <figcaption>, <figure>, <footer>, <form>, <h1>-<h6>, <header>, <hr>, , <main>, <nav>, <noscript>, , <output>, <p>, <pre>, <section>, <table>, <tfoot>, and <video>.

```
<html>
  <head>
  </head>
<body>
  <div style="background-color: lightblue">This is first div</div>
  <div style="background-color: lightgreen">This is second div</div>
  <p style="background-color: pink">This is a block level element</p>
</body>
```

</html>



In the above example we have used

tag, which defines a section in a web page, and takes full width of page.

We have used style attribute which is used to styling the HTML content, and the background color are showing that it's a block level element.

Inline elements:

- Inline elements are those elements, which differentiate the part of a given text and provide it a particular function.
- These elements does not start with new line and take width as per requirement.
- The Inline elements are mostly used with other elements.

<a>, <abbr>, <acronym>, , <bdo>, <big>,
, <button>, <cite>, <code>, <dfn>, , <i>, , <input>, <kbd>, <label>, <map>, <object>, <q>, <samp>, <script>, <select>, <small>, , , <sub>, <sup>, <textarea>, <time>, <tt>, <var>.

<html>

<head>

</head>

<body>

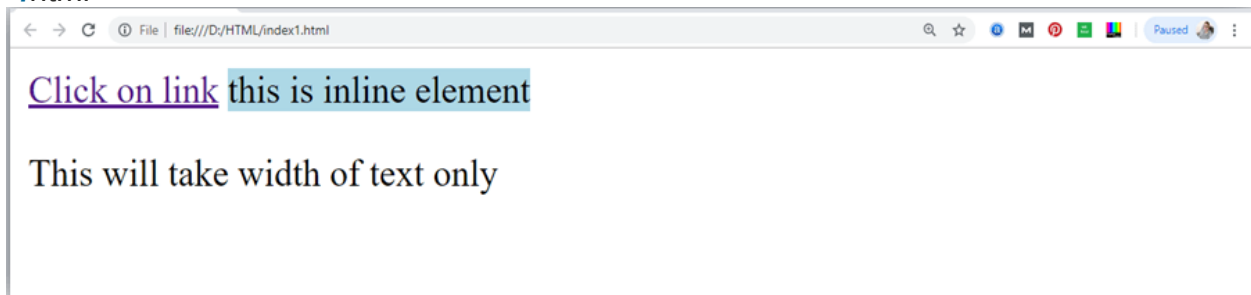
Click on link

this is inline element

<p>This will take width of text only</p>

</body>

</html>



Advantages of HTML 5

- HTML 5 allows you to develop websites in a simplified manner.
- Various HTML headings help the screen readers to access the content in a simplified manner.
- You can add video and audio support in your website by using various tags such as <object> and <embed>.
- Today, HTML 5 is considered to be an important and a relevant language for the Web development industry.
- The HTML 5 standards have simplified the website development for the Web developers and designers.
- HTML 5 allows you to implement client-side storage by implementing JavaScript.
- HTML 5 helps in handling data entry on a website in a better manner.
- You can develop cross-platform games using HTML 5.
- HTML 5 allows you to develop applications for mobile devices.
- Its new tags allow you to implement the drag and drop feature on the website. In other words, the items can be dragged and dropped from one location to another on the same Web page.
- You can create your own vocabularies beyond HTML 5 and extend your Web pages with custom semantics.
- It allows you to implement the functionality of geolocation on a Web page

- Multimedia support
- Short and simple syntax
- Improved security features
- Include semantic tags
- Cross-platform support

Disadvantages

- Client-side rendering
- Local storage is less secure
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- Different video supports for different browsers
- Media licensing cost
- Doesn't support old browsers
- Tough to handle responsiveness on the range of devices

HTML5 is a better version of HTML by improvising HTML and providing features like multimedia compatibility or multi-platform compatibility

Difference between HTML and HTML5?

HTML5 is more complete and easier than **HTML4**, it has lots of new tags like <header>, <footer>, <nav>, <Audio>, <video>, <main> etc. It also supports graphics. In the following image, we have described all the essential terms related to HTML and HTML5.

HTML is referred to as the **primary** language of the **World Wide Web**. **HTML** has many updates over time, and the latest **HTML** version is **HTML5**. There are some differences between the two versions:

- HTML5 supports both **audio** and **video** while none of them were part of
 - HTML cannot allow JavaScript to run within the web browser, while **HTML5** provides full support for running JavaScript.
 - In **HTML5**, inline **mathML** and **SVG** can be used in a text, while in HTML it is not possible.
 - HTML5 supports new types of form controls, such as **date** and **time**, **email**, **number**, **category**, **title**, **Url**, **search**, etc.
 - Many elements have been introduced in HTML5. Some of the most important are **time**, **audio**, **description**, **embed**, **fig**, **shape**, **footer**, **article**,
-

○ Difference between Html and Html5

Features	Html	Html5
definition	A hypertext markup language (HTML) is the primary language for developing web pages.	HTML5 is a new version of HTML with new functionalities with markup language with Internet technologies.
Multimedia support	Language in HTML does not have support for video and audio.	HTML5 supports both video and audio.
Storage	The HTML browser uses cache memory as temporary storage.	HTML5 has the storage options like: application cache, SQL database, and web storage.
Browser compatibility	HTML is compatible with almost all browsers because it has been present for a long time, and the browser made modifications to support all the features.	In HTML5, we have many new tags, elements, and some tags that have been removed/modified , so only some browsers are fully compatible with HTML5 .
Graphics support	In HTML, vector graphics are possible with tools Like Silver light, Adobe Flash, VML , etc.	In HTML5, vector graphics are supported by default.
Threading	In HTML, the browser interface and JavaScript running in the same thread.	The HTML5 has the JavaScript Web Worker API, which allows the browser interface to run in multiple threads.

Storage	Uses cookies to store data.	Uses local storage instead of cookies
Vector and Graphics	Vector graphics are possible with the help of technologies like VML, Silverlight, Flash,etc.	Vector graphics is an integral part of HTML5, SVG and canvas .
Shapes	It is not possible to create shapes like circles, rectangles, triangles.	We can draw shapes like circles, rectangles, triangles